31. The Discriminant Theorem for Orders of an Algebraic Number Field or Function Field

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Dedekind's discriminant theorem says, as is well known: a prime number enters the field discriminant if and only if it is divisible at least by the square of a prime ideal.

I show in what follows that this theorem remains valid for all finite orders of a finite algebraic number field, that is, for all such rings of algebraic integers which contain the ring of all rational integers, in the following form: a prime number p enters the discriminant of an order if and only if, in the decomposition valid in the order of the principal ideal (p) into greatest primary components, at least one proper primary ideal occurs, that is, one which is not a prime ideal.[1]

Since for the principal order, the system of all integers of the field, there are no further primary ideals except powers of prime ideals, Dedekind's theorem follows by specialization.

The proof of the theorem is reduced, by passing to the residue-class ring of the order modulo (p), to the decomposition theory of those rings which are of finite rank with respect to a field contained in the ring, that is, which form a commutative system of hypercomplex quantities with principal identity and coefficients from that same field. Here, in particular, the coefficient field is the residue-class field of the rational integers modulo p, and the decomposition of (p) corresponds in the residue-class ring to the decomposition of the zero ideal.

Thus if such a ring of finite rank is called completely reducible when its zero ideal can be represented as the least common multiple of finitely many prime ideals, the question is to find criteria for complete reducibility. As such a criterion one first obtains that, for a perfect field as coefficient domain, the ring is completely reducible if and only if the extension ring with algebraically closed coefficient domain is completely reducible. For this extension ring, however, the nonvanishing of the discriminant is very easily recognized as a necessary and sufficient criterion for complete reducibility;[2] and since the discriminant of the order modulo each prime number passes into the discriminant of the residue-class ring, the discriminant theorem is thereby proved.[3]

If one considers function fields instead of number fields, the order must now contain the ground domain of all rational integral functions, respectively, for algebraic functions of several indeterminates or for integral algebraic functions, the ground domain of all integral functionals with rational integral coefficients. In that case, the case of an imperfect coefficient domain can also occur in the residue-class ring, for example when, in the case of an integral functional domain, one takes the residue-class ring modulo a prime number. For an imperfect field as coefficient domain one must distinguish between prime ideals of the first and of the second kind according as the residue field modulo the prime ideal is an extension field of the first or of the second kind,[4] and correspondingly between complete reducibility of the first and of the second kind. The criteria stated above refer throughout to complete reducibility of the first kind, which alone occurs in the perfect case. Hence, in the most general case, the discriminant theorem says that a prime element of the ground domain enters the discriminant of an order if and only if, in the ideal decomposition of (p), at least one properly primary component or one prime ideal of the second kind occurs.

For the principal order this fact was first noticed in examples by Kronecker, after the Festschrift had still given, erroneously, the formulation valid for the principal order of a number field, though without a complete proof. For the principal order Ostrowski[5] gave a proof of the above discriminant theorem and further ramification theorems attached to it; at the same time a detailed survey of the literature is found there. Finally, the discriminant theorem for all orders of relative fields follows as a direct consequence, by passing to the quotient ring with respect to each prime ideal of the ground domain. The idea of this passage, which is known in the special case, was emphasized in principle by W. Krull;[6] a systematic theory of the quotient ring in the framework of more general investigations is given by H. Grell.[7]

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[1] Cf. E. Noether, *Abstrakter Aufbau der Idealtheorie in algebraischen Zahl- und Funktionenkörpern*, Math. Ann. 96 (1926), pp. 26–61. For all definitions reference is made to this paper, cited below as "Ideal Theory". In what follows, "order" always means "finite order", and "ring" a commutative ring.

[2] In a special case Hilbert argues similarly: Zur Theorie der aus n Haupteinheiten gebildeten komplexen Größen, Gött. Nachr. 1896.

[3] Related investigations, also in the noncommutative case, but restricted to the principal order (maximal domain) and assuming rational integral number coefficients, are found in the recently published paper of A. Speiser, *Allgemeine Zahlentheorie*, Naturf. Gesellschaft Zürich 71 (1926), pp. 8–48. Cf. also the American literature cited there.

[4] In Steinitz's familiar formulation, J. f. M. 137. Cf. the note to §2, no. 4, of this paper.

[5] A. Ostrowski, Zur arithmetischen Theorie der algebraischen Größen, Nachr. Gött. Ges. d. Wissensch. 1919.

[6] Cf. his Danzig lecture, Jhrbr. d. D. Math. Ver. 34, p. 121.

[7] *Beziehungen zwischen den Idealen verschiedener Ringe* (to appear in Math. Annalen).

The uniform treatment of all orders given here, and of perfect and imperfect coefficient domains, rests, besides the use of ideal theory, on taking from the outset the theory of rings of finite rank, and hence the general concept of a finite extension, as the basis, and not the concept of an extension generated by a primitive element. The module basis of an order, together with the relations among its elements, therefore takes the place of the powers of a primitive element of the field and of the defining equation,[8] respectively of the powers of the fundamental form and of the fundamental equation.

Such a module basis passes, modulo every prime element of the ground domain, into a basis of the residue-class ring, whereas this need not hold even for the principal order when the basis consists of powers of a primitive element; nor for the basis consisting of powers of the fundamental form as soon as integral algebraic functions of several indeterminates are involved.[9] Expressed differently: the residue-class ring of the polynomial domain in one indeterminate with coefficients in the ground domain, modulo an arbitrary defining equation, or even modulo the fundamental equation, need not be isomorphic modulo p to the residue-class ring of the principal order modulo (p). The main difficulties in the customary presentations lie in this disappearance of the isomorphism. Since this auxiliary polynomial domain does not occur here, these difficulties are thereby avoided automatically. At the same time the additive decomposition theory of the residue-class ring can be understood as the equivalent of the multiplicative decomposition of the equation; here too, decomposition into relatively prime factors corresponds to an additive decomposition in the polynomial residue-class ring.

It should be noted that the discriminant theorem can represent only the first theorem of a general ramification theory of orders, just as in the principal order the discriminant theorem is accompanied by the more refined theory of the ramification ideal, which there permits a more precise determination of exponents. This determination of exponents, which rests on the power representation of the primary factors of the equations, can be interpreted as the determination of the length of a composition series from p to $q = p^t$, and will presumably occur in this form in the general ramification theory.

§1. Extension rings of fields

1. We first place in front a few remarks on arbitrary rings. If T is a ring and S a system of elements of T, then by the "ring derived from S in T" is meant the intersection of all rings in T which contain S. Correspondingly one defines the "ideal derived from S in T", or the "P-module derived from S in T", when P is a subring of T.[10]

If R is a subring of T and S a system of elements of T, the ring derived from R and S in T is denoted as the ring $R[S]$ arising by ring-adjunction of S to R. This ring consists of all polynomials in the elements of S with coefficients from R, where the equality and operation relations are fixed by the equality and operation relations in T. Because of these already fixed equality and operation relations it may suffice, in a special case, to form only a subsystem of all polynomials, for instance only all linear forms in S with coefficients from R. Such a special case occurs, for example (cf. no. 2), when R is a ring with identity element and S is a field with the same identity element: all linear combinations $\sum c_i\gamma_i$, where c_i belongs to S and γ_i to R, form the ring $R[S]$.

But when only the ring R is given, and in addition a system S of elements not contained in R with equality relations, or also mutual operation relations, one can also construct the extension ring $R[S]$ arising by adjunction of S to R. One shows that there is always a ring containing R and S, namely the ring formally consisting of all polynomials in S with coefficients from R, with the usual operation rules fixed. For the definition of equality, aside from the operation rules, full freedom is still possible, except for the restriction that the equality definitions valid in R and S must be included. Thus it is necessary only that those polynomials be set equal which, by virtue of equality in R and S and by virtue of the operations, can be formed from equal elements of R and S in the same way. The ring T so constructed, containing R and S, is then identical with the ring derived in T from R and S.[11]

2. Extension rings of fields. From now on assume rings $\mathfrak{R}$ with identity element, containing a field P as a subring, and therefore over-rings of fields. The identity element e of $\mathfrak{R}$ is then also the identity element of P, and the ring is a P-module.

If $\overline{P}$ is an overfield of P, the extension ring $\overline{\mathfrak{R}} = \mathfrak{R}[\overline{P}]$ arising by ring-adjunction of $\overline{P}$ to $\mathfrak{R}$ is to be

[8]This point of view is closely connected with Dedekind's view of higher complex numbers: Zur Theorie der aus n Haupteinheiten gebildeten komplexen Größen, Gött. Nachr. 1885.

[9]Cf. Ostrowski, loc. cit., "irregularity of the first kind".

[10]Cf. Ideal Theory, §1.

[11]Cf. Steinitz's construction of the quotient field.

explained as follows.

It is assumed that the set-theoretic intersection $[\mathfrak{R}, \overline{P}]$ of $\mathfrak{R}$ and $\overline{P}$ is P, and further that between the elements of $\mathfrak{R}$ and $\overline{P}$ not belonging to P no operation relations exist. This assumption can always be achieved by replacing $\overline{P}$ by an equivalent extension field of P, or also by replacing $\mathfrak{R}$ by an equivalent extension ring.[12]

As elements of $\overline{\mathfrak{R}}$ one declares all formally formed linear combinations

$$\bar{c}_{i_1}\gamma_{i_1} + \cdots + \bar{c}_{i_s}\gamma_{i_s},$$

where the $\bar{c}_i$ run through all elements of $\overline{P}$ and the γ_i through all elements of $\mathfrak{R}$.[13] The definition is to be unique; that is, if the γ are replaced by elements equal to them in $\mathfrak{R}$, and the $\bar{c}$ by elements equal to them in $\overline{P}$, then the element in $\overline{\mathfrak{R}}$ is also to be declared equal.

As sum one defines, uniquely in the sense that replacing a summand by an equal one in the equality definition to be fixed gives the same result,

$$\sum \bar{c}_i\gamma_i + \sum \bar{d}_i\gamma_i = \sum(\bar{c}_i + \bar{d}_i)\gamma_i.$$

Here zeros may occur among the $\bar{c}$ and $\bar{d}$, whereby formally the same γ_i occur.

As product one defines, in the same sense uniquely,

$$\sum \bar{c}_i\gamma_i \cdot \sum \bar{d}_k\gamma_k = \sum \bar{c}_i\bar{d}_k\gamma_i\gamma_k.$$

For the equality definition it is declared that, if $\gamma_{i_1}, \ldots, \gamma_{i_s}$ are linearly independent with respect to P, then two elements

$$\bar{c}_{i_1}\gamma_{i_1} + \cdots + \bar{c}_{i_s}\gamma_{i_s} \quad \text{and} \quad \bar{d}_{i_1}\gamma_{i_1} + \cdots + \bar{d}_{i_s}\gamma_{i_s}$$

are equal if and only if $\bar{c}_{i_k}$ is equal to $\bar{d}_{i_k}$ in $\overline{P}$. Expressed differently: elements of $\mathfrak{R}$ linearly independent with respect to P are declared to be linearly independent with respect to $\overline{P}$.

By the definitions of sum and product the equality definition for all elements of $\overline{\mathfrak{R}}$ is thereby given. Indeed their expressibility by linearly independent elements from $\mathfrak{R}$ follows. For from a relation

$$\mu = c_1\gamma_1 + \cdots + c_s\gamma_s \quad \text{in } \mathfrak{R}$$

one obtains, by multiplication with $\bar{b} = be$ according to the product definition,

$$\bar{b}\mu = \bar{b}e \cdot (c_1\gamma_1 + cdots + c_s\gamma_s) = \bar{b}c_1\gamma_1 + \cdots + \bar{b}c_s\gamma_s,$$

and hence, by the sum definition, the expressibility of all elements by linearly independent elements from $\mathfrak{R}$.

Consequently, if all elements have been expressed by linearly independent elements from $\mathfrak{R}$, the equality definition is equality of coefficients in $\overline{P}$. It therefore agrees, for elements belonging simultaneously to $\mathfrak{R}$ and $\overline{\mathfrak{R}}$ or to $\overline{P}$ and $\overline{\mathfrak{R}}$, with the equality valid there; whereas for the remaining elements of $\overline{\mathfrak{R}}$ the equality valid in $\mathfrak{R}$ and $\overline{P}$ says nothing beyond the uniqueness required in the definition, because of the intersection assumption and the further condition.[14] Similarly, the requirement that sum and product be unique entails no further relations, since equal elements may be assumed coefficientwise equal, and hence formally give the same sum and product expressions.

One now verifies at once that all axioms of the ring operation are fulfilled; thus the extension ring $\mathfrak{R}[\overline{P}]$ generated by ring-adjunction always exists.

3. Ideals in $\mathfrak{R}$ and $\overline{\mathfrak{R}}$. If $\mathfrak{a}$ is an ideal in $\mathfrak{R}$, then the module product $\overline{\mathfrak{R}}\mathfrak{a}$ is an ideal $\bar{\mathfrak{a}}$ in $\overline{\mathfrak{R}}$, the extension ideal of $\mathfrak{a}$. At the same time $\bar{\mathfrak{a}}$ is the ideal derived from $\mathfrak{a}$ in $\overline{\mathfrak{R}}$; it consists of all linear combinations $\sum \bar{c}_i a_i$, the sum in each case being over finitely many elements, where the a_i run through all elements of $\mathfrak{a}$ and the $\bar{c}_i$ through all elements of $\overline{P}$.

[12] Two extension fields of P are called "equivalent", following Steinitz, if they can be placed in isomorphism with one another in such a way that P corresponds elementwise to itself. Equivalent extension rings of P are to be defined correspondingly. By introducing new symbols one can always produce equivalent extensions; this freedom in the use of equivalent extensions is essential for what follows.

[13] In general, elements from $\mathfrak{R}$ or $\overline{\mathfrak{R}}$ are denoted by Greek letters, and elements from P or $\overline{P}$ by Latin letters.

[14] Without this assumption there would be at least one element $\bar{c}$ belonging to $\overline{P}$ but not to P, for which $\bar{c} = \bar{c}e = \gamma$ would hold. Then, by the equality definition in $\mathfrak{R}$, e and γ would indeed be linearly independent with respect to P, but dependent with respect to $\overline{P}$. Because of the relations between elements of $\overline{P}$ and $\mathfrak{R}$, the above equality definition would then not be possible; think, for instance, of the lowering of degree of a finite extension field when the ground field is replaced by an intermediate field. The above extension is, by contrast, characterized by the possibility of the appearance of new zero divisors. Thus, for example, the residue-class ring of the polynomial domain with rational coefficients P modulo x^2+1 is a ring without zero divisors, isomorphic to the field $P(\sqrt{-1})$, whereas on passing to the polynomial domain with coefficients from $P(\sqrt{-1})$ zero divisors occur, corresponding to $(x+i)(x-i) = x^2+1$, although no factor is divisible by x^2+1. The corresponding statement holds, for example, for every finite number field; this is the conception first expressed by Dedekind for the higher complex numbers. Cf. on this point §6, no. 4, at the end.

If $\overline{\mathfrak{b}}$ is an ideal in $\overline{\mathfrak{R}}$, then the intersection $[\overline{\mathfrak{b}}, \mathfrak{R}]$ is an ideal $\mathfrak{b}$ in $\mathfrak{R}$, the contraction ideal of $\overline{\mathfrak{b}}$.[15] Every ideal $\mathfrak{a}$ of $\mathfrak{R}$ is a contraction ideal, namely of its extension ideal $\overline{\mathfrak{a}}$. Indeed, every element a of $[\mathfrak{R}, \overline{\mathfrak{a}}]$ is of the form $\sum \bar{c}_i a_i$, where the a_i may be assumed linearly independent with respect to P or $\overline{P}$. Thus the elements a, a_i become linearly dependent in $\overline{\mathfrak{R}}$, and hence, by the equality definition, also in $\mathfrak{R}$; therefore a belongs to $\mathfrak{a}$. Since the converse also holds, $\mathfrak{a} = [\mathfrak{R}, \overline{\mathfrak{a}}]$.

If $\overline{\mathfrak{p}}$ is a prime ideal in $\overline{\mathfrak{R}}$, then $\mathfrak{p} = [\mathfrak{R}, \overline{\mathfrak{p}}]$ is a prime ideal in $\mathfrak{R}$. For, by the second isomorphism theorem,[16] the residue-class ring $\mathfrak{R}/\mathfrak{p}$ is isomorphic to a subring of $\overline{\mathfrak{R}}/\overline{\mathfrak{p}}$, and hence is a ring without zero divisors. Namely, there is the ring isomorphism

$$(\mathfrak{R}, \overline{\mathfrak{p}})/\overline{\mathfrak{p}} \simeq \mathfrak{R}/[\mathfrak{R}, \overline{\mathfrak{p}}],$$

where $(\mathfrak{R}, \overline{\mathfrak{p}})$ denotes the sum, the greatest common divisor, of the modules $\mathfrak{R}$ and $\overline{\mathfrak{p}}$.

§2. Rings of finite rank. Representation as a direct sum

1. Rings of finite rank. If $\mathfrak{R}$ is a finite P-module, then $\mathfrak{R}$ is of finite rank, say k, with respect to P: that is, there are k, and no more, elements in $\mathfrak{R}$ linearly independent with respect to P, and every system of k elements of $\mathfrak{R}$ linearly independent with respect to P consequently forms a module basis of $\mathfrak{R}$. If a P-module $\mathfrak{B}$ of finite rank is divisible by another $\mathfrak{A}$ of the same finite rank, $\mathfrak{B} = O(\mathfrak{A})$, then they are therefore identical.

From now on assume that $\mathfrak{R}$ has finite rank n with respect to P. Then every P-module in $\mathfrak{R}$ has finite rank. Thus in every proper chain of divisors or multiples of P-modules in $\mathfrak{R}$, the rank of each module must be larger, respectively smaller, than the rank of the immediately preceding one; the chains break off after finitely many steps. In particular, the "double-chain theorem" holds for the system of all ideals in $\mathfrak{R}$.

It follows from this[17] that a prime ideal $\mathfrak{p}$ of $\mathfrak{R}$ has no proper divisor different from the identity ideal; hence the residue-class ring $\mathfrak{R}/\mathfrak{p}$ modulo a prime ideal different from the identity ideal is a field. If $\mathfrak{q}$ is a primary ideal belonging to $\mathfrak{p}$, that is, $\mathfrak{q} = O(\mathfrak{p})$ and $\mathfrak{p}^\rho = O(\mathfrak{q})$, then the residue-class ring $\mathfrak{R}/\mathfrak{q}$ is a primary ring, that is, a ring in which a power, here in any case already the ρ-th, of every zero divisor vanishes; and $\mathfrak{R}/\mathfrak{q}$ contains only zero divisors and units. The latter follows immediately from the fact that every ideal not divisible by $\mathfrak{p}$ is relatively prime to $\mathfrak{p}$, and hence also to $\mathfrak{q}$.

From the double-chain theorem and from the existence of the identity element it follows further[18] that every ideal $\mathfrak{a}$ of $\mathfrak{R}$ can be represented uniquely as a product of finitely many pairwise relatively prime primary ideals. Because of relative primeness this product is equal to the least common multiple and corresponds to a representation of the residue-class ring $\mathfrak{R}/\mathfrak{a}$ as a direct sum of primary rings.[19] That is, $\mathfrak{R}/\mathfrak{a}$ is the greatest common divisor of these rings, and the sum representation of every element of $\mathfrak{R}/\mathfrak{a}$ is unique.

2. Representation of rings of finite rank as direct sums of primary rings. Suppose that the representation of the zero ideal of $\mathfrak{R}$ is given by

$$(0) = \mathfrak{q}_1\mathfrak{q}_2 \cdots \mathfrak{q}_r = [\mathfrak{q}_1, \mathfrak{q}_2, \ldots, \mathfrak{q}_r],$$

and let the corresponding representation as a direct sum be

$$\mathfrak{R} = \mathfrak{R}_1 + \mathfrak{R}_2 + \cdots + \mathfrak{R}_r, \qquad \mathfrak{R}_i\mathfrak{R}_k = (0)\ (i \neq k), \qquad \mathfrak{R}_i^2 = \mathfrak{R}_i, \qquad \mathfrak{R}\mathfrak{R}_i = \mathfrak{R}_i.$$

Then $\mathfrak{R}_i$ is an ideal of $\mathfrak{R}$, namely

$$\mathfrak{R}_i = \mathfrak{q}_1 \cdots \mathfrak{q}_{i-1}\mathfrak{q}_{i+1} \cdots \mathfrak{q}_r = [\mathfrak{q}_1, \ldots, \mathfrak{q}_{i-1}, \mathfrak{q}_{i+1}, \ldots, \mathfrak{q}_r],$$

$$\mathfrak{q}_i = \mathfrak{R}_1 + \cdots + \mathfrak{R}_{i-1} + \mathfrak{R}_{i+1} + \cdots + \mathfrak{R}_r,$$

and $\mathfrak{R}_i$ is isomorphic to the residue-class ring $\mathfrak{R}/\mathfrak{q}_i$: to each element γ_i of $\mathfrak{R}_i$ corresponds the class represented by γ_i in $\mathfrak{R}/\mathfrak{q}_i$.[20] If $\mathfrak{a}$ is any ideal of $\mathfrak{R}$, then the distributive law gives

$$\mathfrak{a} = \mathfrak{R}\mathfrak{a} = \mathfrak{R}_1\mathfrak{a} + \cdots + \mathfrak{R}_r\mathfrak{a} = \mathfrak{a}_1 + \cdots + \mathfrak{a}_r;$$

[15] In H. Grell's notation, in the work cited in note 6 on p. 83 of the original.

[16] Ideal Theory, §4, no. 3. The theorem is there stated only for ideals of the same ring; the same consideration shows its validity in the above form for arbitrary ideals $\overline{\mathfrak{a}}$ of $\overline{\mathfrak{R}}$. The ring homomorphism $\overline{\mathfrak{R}} \to \overline{\mathfrak{R}}/\overline{\mathfrak{a}}$ induces, for the subring $\mathfrak{R}$ of $\overline{\mathfrak{R}}$, the ring homomorphism $\mathfrak{R} \sim (\mathfrak{R}, \overline{\mathfrak{a}})/\overline{\mathfrak{a}}$. Under this homomorphism all and only the elements of $[\mathfrak{R}, \overline{\mathfrak{a}}]$ correspond to the zero element, and hence the ring isomorphism $(\mathfrak{R}, \overline{\mathfrak{a}})/\overline{\mathfrak{a}} \simeq \mathfrak{R}/[\mathfrak{R}, \overline{\mathfrak{a}}]$ follows. The classes can be represented on the right and on the left by the same elements of $\mathfrak{R}$, for at each step of the proof the elements are assigned to the classes represented by them.

[17] Ideal Theory, §7, no. 2.

[18] Ideal Theory, §7, no. 3, Theorem V.

[19] Ideal Theory, §4, no. 5. What is added in the present compilation is found scattered in many places in the literature.

[20] Ideal Theory, §4, no. 5.

here, since $\mathfrak{a}_i = \mathfrak{R}_i\mathfrak{a} = \mathfrak{R}\mathfrak{a}_i$, the components $\mathfrak{a}_i$ are ideals both in $\mathfrak{R}_i$ and in $\mathfrak{R}$. As ideals, the $\mathfrak{a}_i$, like the $\mathfrak{R}_i$ themselves, are finite P-modules; because the sum is direct, their ranks add to the rank of $\mathfrak{a}$, respectively of $\mathfrak{R}$.

If the sum representation of the identity element is given by

$$e = \varepsilon_1 + \cdots + \varepsilon_r, \qquad \varepsilon_i\varepsilon_k = 0\ (i \neq k), \qquad \varepsilon_i^2 = \varepsilon_i, \qquad \varepsilon_i \equiv e \pmod{\mathfrak{q}_i},$$

then the representation of every element γ of $\mathfrak{R}$ is obtained uniquely as

$$\gamma = \gamma e = \gamma\varepsilon_1 + \cdots + \gamma\varepsilon_r = \gamma_1 + \cdots + \gamma_r,$$

where the component $\gamma_i = \gamma\varepsilon_i$ is an element of $\mathfrak{R}_i$; at the same time $\mathfrak{R}_i = \mathfrak{R}\varepsilon_i$. From the "orthogonality relations" one obtains

$$\gamma\delta = (\gamma_1\varepsilon_1 + \cdots + \gamma_r\varepsilon_r)(\delta_1\varepsilon_1 + \cdots + \delta_r\varepsilon_r) = \gamma_1\delta_1\varepsilon_1 + \cdots + \gamma_r\delta_r\varepsilon_r.$$

Thus $\gamma_i\delta_i$ is the i-th component of the product, and similarly $\gamma_i + \delta_i$ is the i-th component of the sum, facts which also follow from the homomorphism from $\mathfrak{R}$ to $\mathfrak{R}_i$.

The ring $\mathfrak{R}_i$ contains a subfield $P_i = P\varepsilon_i$ isomorphic to P and is of finite rank with respect to P_i; this rank is equal to the rank possessed by $\mathfrak{R}_i$ as a P-module. For for every element c of P, one has $c\varepsilon_i \equiv ce \equiv c \pmod{\mathfrak{q}_i}$, and hence $c\varepsilon_i \neq 0$ as soon as $c \neq 0$; the homomorphism between P and P_i becomes an isomorphism. Further, every linear dependence with respect to P of elements of $\mathfrak{R}_i$ passes, by multiplication with ε_i, into such a dependence with respect to P_i; conversely, because $P_i = P\varepsilon_i$, every relation with respect to P_i is at the same time a relation with respect to P. Thus the ranks with respect to P and P_i agree.

3. Direct sum representation of extension rings. If

$$\mathfrak{R} = \mathfrak{R}_1 + \cdots + \mathfrak{R}_r,$$

then for $\overline{\mathfrak{R}}$ a representation

$$\overline{\mathfrak{R}} = \overline{\mathfrak{R}}_1 + \cdots + \overline{\mathfrak{R}}_r$$

holds, where $\overline{\mathfrak{R}}_i$ denotes the extension ideal of $\mathfrak{R}_i$ in $\overline{\mathfrak{R}}$, at the same time the ring derived from $\mathfrak{R}_i$ in $\overline{\mathfrak{R}}$. Indeed, from $(0) = \mathfrak{q}_1 \cdots \mathfrak{q}_r$, multiplication by $\overline{\mathfrak{R}}$ gives the product representation $(0) = \overline{\mathfrak{q}}_1 \cdots \overline{\mathfrak{q}}_r$, where the $\overline{\mathfrak{q}}_i$ are again pairwise relatively prime. Hence one obtains a direct-sum representation for $\overline{\mathfrak{R}}$, and since the i-th component of this representation is given by

$$\overline{\mathfrak{q}}_1 \cdots \overline{\mathfrak{q}}_{i-1}\,\overline{\mathfrak{q}}_{i+1} \cdots \overline{\mathfrak{q}}_r,$$

it is equal to the extension ideal of $\mathfrak{R}_i$. But by §1, no. 3, this extension ideal is identical with the extension ring $\mathfrak{R}_i[\overline{P}_i]$ formed from $\mathfrak{R}_i$ with respect to $\overline{P}_i = \overline{P}\varepsilon_i$; therefore in extensions it suffices to extend the individual components. In general $\overline{\mathfrak{R}}_i$ will then no longer be a primary ring.

4. Prime ideals of the first and second kind. By no. 1, the residue-class ring $\mathfrak{R}/\mathfrak{p}$ modulo every prime ideal different from the identity ideal is a field. This field possesses a subfield $(P) = (P, \mathfrak{p})/\mathfrak{p} = P/[P, \mathfrak{p}] \simeq P$ isomorphic to P, since $\mathfrak{p}$ can contain no nonzero element from P; hence (P) consists of all and only the classes which can be represented by elements of P.[21] Thus $\mathfrak{R}/\mathfrak{p}$ is of finite rank with respect to (P), hence is a finite algebraic extension field. According as this extension is of the first or second kind,[22] $\mathfrak{p}$ is to be called a prime ideal of the first or of the second kind.

[21] Cf. the note to §1, no. 3, on the second isomorphism theorem.

[22] In the familiar terminology of Steinitz: an algebraic extension is called of the first kind if every element is a zero of a prime function which, in a suitable extension field, decomposes into distinct linear factors; otherwise it is of the second kind.